MALMÖ UNIVERSITY

Reimagining waste

Upcycling Plastic Waste in Product Design

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Abstract

This project explores how designers and consumers can implement design for sustainability to create products using plastic waste - facilitating proper disposal practices and ultimately contributing to the transition toward a circular economy. This study highlights the importance of innovative and sustainable design that contributes to a circular economy by examining how design principles can be implemented to transform and repurpose discarded materials. The study uses theories on design for sustainability (DFS), circular economy, circular design, and upcycling. Practical design methods that were conducted included material-driven design (MDD), interviews, future scenarios, sketching, and prototyping. This included hands-on exploration and researching the material's inherent qualities, the material in question, and ABS plastic. Expert interviews and onboard research uncovered that most plastic waste ends up in landfills, and recycled plastic, in truth, goes to incineration, renamed energy recycling. MDD revealed that when working with ABS, the temperature affects the material's thickness; higher temperatures, such as 170 Celsius, yield thinner thickness, and temperatures below 120 Celsius result in thicker. This resulted in a design concept that uses upcycling of scrap from ABS plastic and heating it to form panels that later turned into parts for lamps. This method proves to be a feasible and effective method to upcycle plastic waste with a low threshold for designers and consumers to conduct at home or in a studio.

Keywords: DFS, MDD, ABS,

Wordlist

DFS: Design for Sustainability
MDD: Material-Driven Design
ABS: Acrylonitrile Butadiene Styrene
Vitrimers: A class of renewable plastics derived from thermosetting polymers that exhibit high strength, durability, and chemical resistance
End-of-life (EOL): The last life cycle stage of a product or material

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1. Introduction

Plastic is a common material and a collective name for various polymer compounds (Thompson, 2017). Its technical ability to be transformed into several shapes and its cheap production make plastic desirable, from simple packaging to complex parts for aircraft. Plastic contributes to satisfying personal and societal needs. However, today's use of plastic could be more sustainable. Although Sweden has a significant incentive to sort waste at source, it is mainly used for incineration. Eighty-seven percent of plastic waste is estimated to go to energy recovery through incineration, while just over 10 percent is recycled into new products (Naturvårdsverket, 2023). On the contrary, if we observe how the rest of the world disposes of plastic waste, 9 percent is recycled, 12 percent is incinerated, and 79 percent accumulates in landfills and oceans (Balu, Dutta & Choudhury, 2022). The consequence of landfilling is the leaching of toxic chemicals into the ground and the water, impacting society and nature. This project will explore how designers and consumers interested in recycling can create products from plastic waste using upcycling,

1.1 Problem Statement

How can end-of-life material from plastic waste be used to create products within product design?

1.2 Purpose of study

This study aims to explore the concept of upcycling within the realm of design, focusing on how designers and consumers can transform plastic waste into valuable resources or products in a sustainable manner. This study aims to highlight the importance of innovative and sustainable design that contributes to a circular economy by examining how design principles can be implemented to transform and repurpose discarded materials. This study seeks to transition toward responsible and conscious design practices by analyzing the impact of upcycling on product development and sustainability. Such practices benefit the environment and fit the circular economy.

1.3Delimitations

The project will be limited to only one type of plastic to prevent mixing materials, as the project has sustainable development in mind while reducing uneven structures and properties that can occur when plastics are mixed.

1.4 Ethical limitation

This project will not involve animals or people in implementing the method except for interviews. If the interviewee wishes to remain anonymous, they are given the right to do so and can end the interview at any point. Audio, images, and videos are not distributed on different platforms, but interview transcripts may be used for references in the project.

2. Background

Plastic has become a mundane part of modern life, with its wide range of applications. However, the persistent issue of plastic waste and the challenges associated with its management sparks global concerns. A significant problem is the lack of proper standardized identification for petroleum-based plastic products, making it difficult to effectively identify, sort, and recycle different types of plastic in mixed waste streams. Although there are more than 700 varieties of petroleum plastic, only six are correctly labeled with an individual code number; meanwhile, the rest are identified with code number 7, labeled as “other” (Sörensen, 2021). This results in an excessive increase in landfills, causing soil and water contamination, hindering the efforts to reduce plastic pollution and promoting a circular economy. Ultimately, addressing the issue of inadequate identification is crucial for enhancing the sustainability of plastic use and improving waste management practices.

Acrylonitrile Butadiene Styrene (ABS) is a versatile, impact-resistant thermoplastic polymer known for its strength, rigidity, and wide color range (Thompson, 2017). ABS plastic is lightweight, has low manufacturing costs, and can be easily molded or extruded through blow and injection molding. It can produce highly detailed items for electronics and oversized parts such as car bumpers, making it desirable for manufacturers. It is typically utilized in various applications, including household items, toys, automotive parts, housing for electronics, and other enclosures. ABS is fully recyclable, promoting a more sustainable approach to using plastic; however, despite its beneficial characteristics, ABS is not biodegradable and is often improperly recycled or disposed of in landfills or incinerators because it is identified with plastic code 7 “other,” making it challenging to separate from mixed waste. This project has opted to work with ABS plastic since it is a standard plastic used in a wide range of consumer products compared to other plastics with code number 7, making it an abundant resource that can be collected. Due to its recyclability, it allows for products with longer life cycles before incineration if they are contained in a cycle. The project will focus on how designers and consumers can upcycle ABS from post-consumer waste. This project highlights the importance of promoting proper recycling methods to ensure the long-term sustainability of materials such as ABS and other plastics labeled “other.”

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The inspiration for the project has been personal previous work on design for sustainability, in addition to personal preferences such as Scandinavian design and Buddhistic architectures such as pagodas that have affected the result of the design.

Previous work by Balu et al. (2022) researched sustainable solutions for waste management through upcycling plastic waste. The article presents recent advances in upcycling plastic waste and examines emerging methods. It includes vitrimerization, fabrication of nanocomposites, additive manufacturing, catalytic conversion, and industrial biotechnology. The conclusion they reached is that vitrimerization and fabrication of nanocomposites can potentially be applied for future production because the advantage of these methods is that they result in a solid structure and mechanical properties, as well as that they can be recycled without losing their strength in the material compared to previous products. The article concluded that the advantage of additive manufacturing, catalytic conversion, and industrial biotechnology is that they can create diverse products such as allotropes, fuel, and degradable products.

Past studies researching recycling upcycling ABS include Zhao et al. (2022), who created ABS pellets from plastic waste for 3D printing filaments. Nam et al.(2023) researched how to upcycle ABS plastic from post-consumer waste with the advanced process of using plasma-assisted mechanochemistry (PMC), resulting in higher recycling efficiency than commercial ABS in addition to improved mechanical properties due to the mixture of styrene-butadiene-styrene (SBS).

Kim et al. (2022) studied how to upcycle closed-loop ABS plastic through vitrimerization, turning regular ABS into ABS-vitrimer. Similar to the study of Balu et al. (2022), the study of Kim et al. displayed that ABS-vitrimers facilitate circularity without losing the technical properties of the material and are separable from unsorted plastic waste.

3 Theory

3.1 Design For Sustainability

Sustainable development aims to provide today's needs sustainably without compromising future needs (Kates, Parris & Leiserowitz, 2005). The European Commission (n.d) has legislated the Plastics Strategy, which aims to transition toward a carbon-neutral and circular economy by 2030. This strategy is mentioned as a key to reaching the sustainable development goals, the Paris Climate Agreement, and the EU's industrial policy objectives by implementing regulations on packaging and increasing the demand for recycled plastic content (European Commission, n.d). Tackling this challenge of balancing society's necessities without infringing on future generations requires innovative design practices that are conscious of the impact on society and the environment throughout the process. Design for sustainability prioritizes human and environmental well-being by designing products with a positive impact, considering their life cycle and ecological effects (Ceschin & Gaziulusoy, 2020). It emphasizes resource efficiency, the use of eco-friendly materials, and awareness of the environmental, climate, and ecosystem impacts of products.

3.2 Circular Economy

A circular economy is described as an economy where waste does not occur, but resources can be retained in society's cycle or returned sustainably to nature's cycle (Regeringen, 2017). European Parliament (2023) defined a circular economy as a production and consumption model involving sharing, leasing, reusing, repairing, refurbishing, and recycling existing resources and products for as long as possible, in contrast to a traditional linear economic model, which is purely based on consumption followed by a waste pattern. In practice, a circular economy means minimizing waste by recycling materials from products at the end of their life cycle to keep them in circulation within the economy, which can be achieved with recycling. Figure 1 from the Ellen Macarthur Foundation (2019), exhibits a simplified model that illustrates how products are processed in a circular economy. According to the European Parliament, a circular economy allows for the repeated and productive utilization of materials, ultimately generating additional value. Reusing and recycling products and materials slows the use of finite natural resources, reduces landscape and habitat disruption, and helps limit biodiversity loss.

3.3 Circular Design

Circular design is a sustainable development method based on the circular economy (Balkenende, Bakker & Bocken 2018). The circular design strives for products to remain in a cycle and has as its starting point to design products to minimize their environmental impact without losing their economic value for as long as possible. It is achieved by designing and extending the life cycle of products and by enabling efficient repair, refurbishment, remanufacturing, and recycling to reuse products, components, and materials in the economic system.

3.4 Upcycling

Upcycling is described as a methodical way of enhancing the value of waste material through creative reutilization, making traditional recycling methods obsolete (Richardson, 2011). While recycling efforts can achieve sustainable solutions, conserving energy and water is more effective by circumventing the need to reprocess material to its virgin state. The challenge with upcycling is that recycling can downgrade the material and decrease the inhabited qualities. Upcycling focuses on integrating discarded components into innovative products through open-loop cycles rather than remanufacturing, where products are refurbished within the same closed-loop system. Upcycling materials creatively can transform items into various other products, maintaining or even increasing their value. Designers need to consider the whole life cycle of the products to ensure that they can be repurposed.

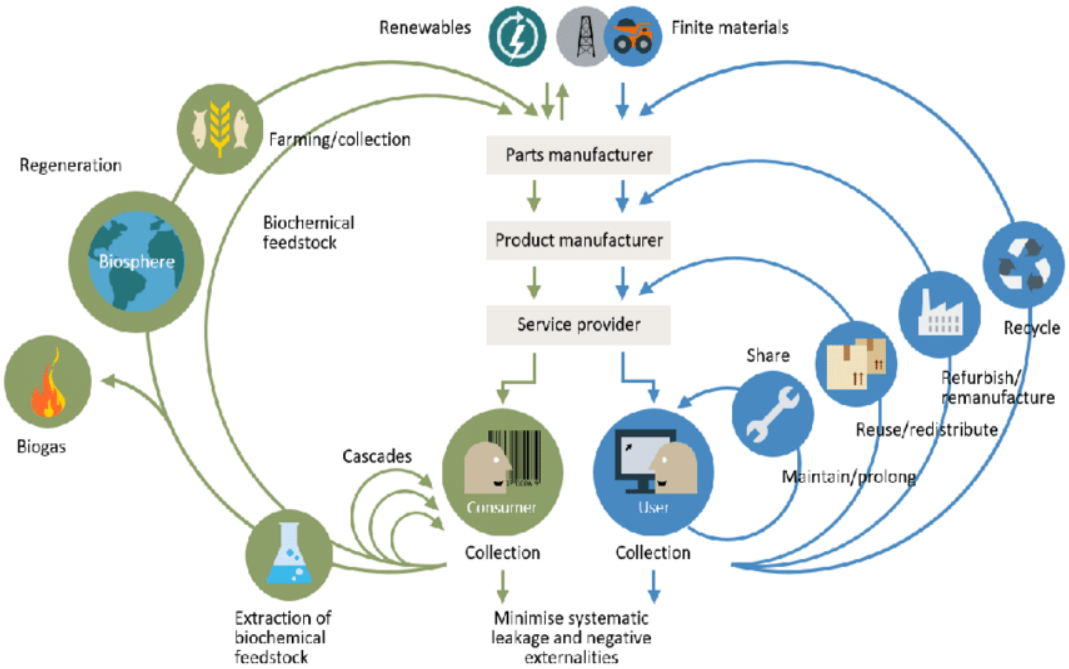


Fig 1: Butterfly diagram. Ellen Macarthur Foundation (2019)

3.5 Speculative design

Speculative design is described as a methodology and strategy that merges emerging technology and techniques borrowed from film, literature, ecology, comedy, and psychology to develop and present plausible futures (Auger, 2013). Speculative design shares similarities with other design methods such as Critical Design, Discursive design, Design probes, and Design fiction as these methods use fiction to present alternative products, systems, or worlds. Auger (2013) states that speculative design needs a plausible future scenario to relate to the audience and loses its immersion if it strays too far from the audience's habits and reality. Therefore, speculative design requires a bridge that connects the audience's perception of the world and the fictional elements of the design concept to be convincing and engaging.

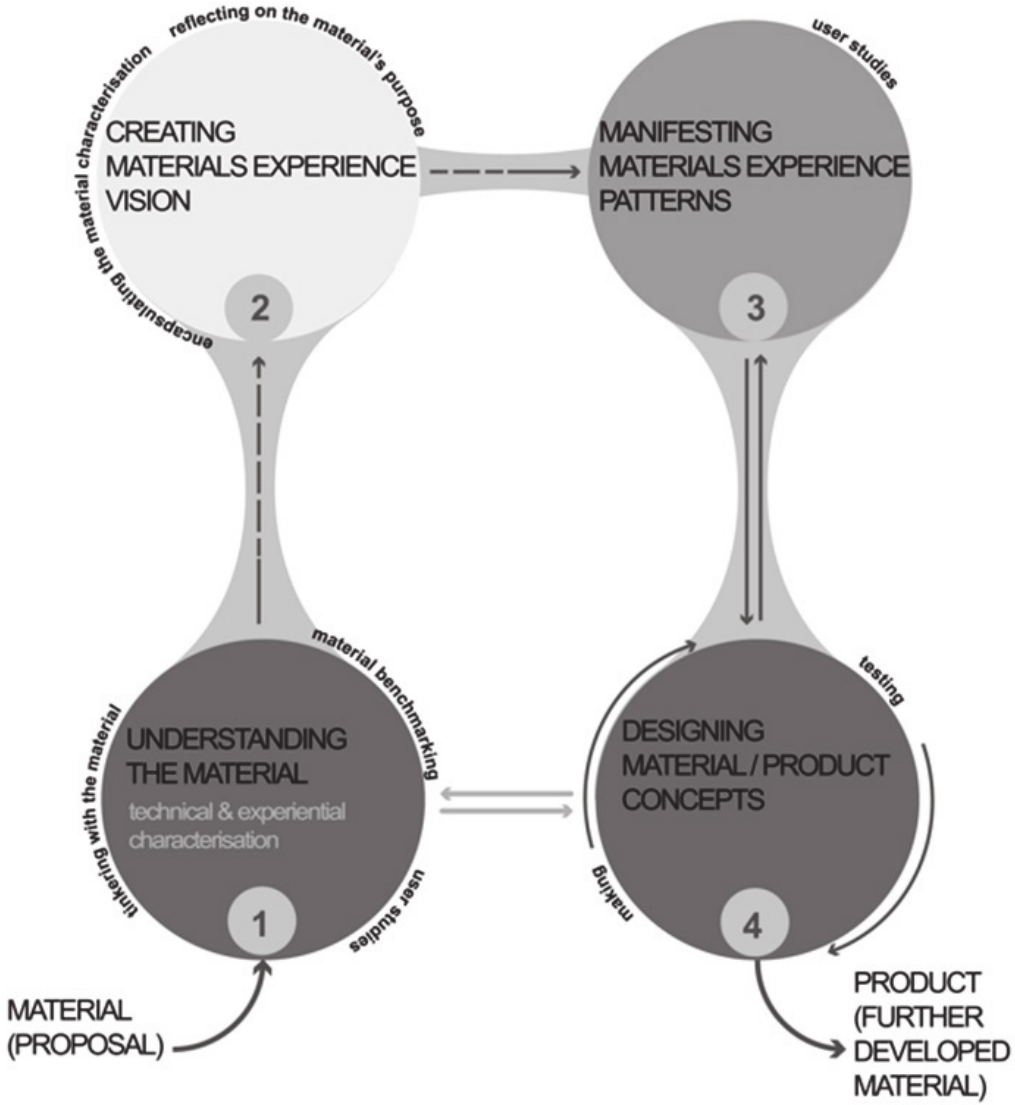


Fig 2. Material Driven Design (MDD) method. Karana et al. (2015)

4. Methods

4.1 Material-Driven Design

Material-driven design is a method that welcomes exploration and understanding of the material's inherent qualities and potential in the design process (Karana, 2015). Designers can create solutions not only for function or form but also for the material's inherent properties by valuing the properties of the materials. Material-driven design demonstrates an abstract approach through hands-on explorations and harnessing materials to facilitate the creative process of idea generation instead of presenting problems through rapport (van Bezooyen, 2018). Karana et al. (2015) state that several designers have started implementing material-driven designs to explore various textures and surfaces for potential materials.

Karana's (2015) material-driven design follows a 4 step process for designing a product concept. The first step involves understanding the material in question to know the inherent qualities and limits of the material, simultaneously comparing it to material in the same field. This involves material understanding, material tinkering, and material benchmarking. The second step of MDD is to create a vision of materials experience, which helps designers summarize observations and numerous findings. Karana (2015) explains that materials experience vision expresses the designer's envision of the material's role in creating or designing a product. The third step of MDD uses the vision from step 2 to develop an interrelationship between the material experience vision and the formal qualities of materials and products. Here, the designers must reflect on how other people will experience and interact with the material in a way they envision rather than speculative interaction or intuitions. The fourth step of MDD is to develop product concepts based on the previous steps.

This project follows Karana's model in Figure 2 to a certain degree and uses step 1 of Karana's model, following material understanding, material tinkering, and material benchmarking. After conducting step 1 of MDD, the project will use the insights from step 1 of MDD to conduct expert interviews and develop future scenarios instead of continuing with steps 2 and 3. This decision was made because the project objective is to explore the potential of plastic waste in product design. By emphasizing future scenarios and interviews, the project will gain a deeper understanding of how plastic waste can be repurposed into new products and further promote new recycling methods and circular designs.

4.1.1 Material understanding

The initial step in material-driven design is understanding the material's unique characteristics and limitations while comparing it with competing materials in similar applications. This process was conducted by reading through The Material Sourcebook For Design Professionals (2017), reading online research, and physically examining the material firsthand. By understanding the material's properties, nature, and constraints more deeply, designers can effectively implement their potential to create innovative and functional products.

4.1.2 Material tinkering

The tinkering process was conducted to explore the possibilities of ABS plastic and push the boundaries of the material. The samples gave insight into how ABS behaved and interacted. Figure 3 highlights how material samples were created by gathering granules, scraps of ABS, a contact grill, non-stick grill mesh, a spatula, a scale, and a fork. The tinkering process was performed in a well-ventilated area with a fume extractor to remove harmful chemicals in the air.

The contact grill was set to preheat at 120 Celsius for 5 minutes with the grill mesh.

Twenty grams of granules and scrap pieces were piled up on the grill mesh and pressed down on the grill for 15 minutes.

The samples were later removed from the grill and placed on a surface to cool.

The procedure was repeated with higher temperatures, from 140, 160 to 170 Celsius.



Fig 3. Scraps of ABS. Louis Diep (2024)

4.1.3 Material benchmarking

Following the tinkering of the material, the samples were benchmarked to assess their inherent qualities. This was tested by sawing, hammering, scratching, dropping, and molding. The molding of the sample involved putting the sample in the contact grill for 5 minutes at a temperature of 170 Celsius.

4.2 Expert interview

Expert interviews are structured or semi-structured dialogues that provide qualitative information from experts in a particular field (Bogner, Littig, & Menz, 2009). Bogner et al. (2009) explain that expert interviews can be conducted several times during the project. In the initial phase, expert interviews effectively offer insights into information difficult to access, and a deeper understanding of user behavior, preferences, and pain points related to current products. Expert interviews allow for engagement and provide feedback on developed concepts, making it a valuable tool for refining design ideas at the later stages of the design process.

The expert interview was conducted with experts in waste management to gain more profound knowledge of current plastic separation methods, pinpoint the challenges that occur today, assess emerging advancements in plastic recycling, and investigate the expert's vision for the future of plastic and recycling. Questions such as: How is plastic separated today? Is it possible to identify individual types of plastic in a large-scale mixture? If so, how? What kinds of plastic are sorted and recycled? The interview was recorded through a Zoom meeting. After the interview was finished, the audio was transcribed.

4.3 Future Scenario

Future scenarios are methodical ways of involving research, imagination, and creativity to anticipate potential outcomes that may occur in the future (Arruda, 2021). Designers can plan for various contingencies and address future challenges by speculating and constructing different scenarios. This method allows designers to explore innovative solutions, reassess assumptions, and ultimately design a product. Future scenarios enable designers to think critically about the impact of their design and product by reflecting on how their design works in society and the environment. Figure 4 illustrates the aim of future scenarios by showcasing how the present has to think and speculate on potential situations to reach a desired or preferable future. Future scenarios were incorporated into this project to address the current environmental situation and consider approaching plastic and waste management more sustainably; one method is upcycling. By speculating on potential societal shifts toward better waste management practices, promoting the design of products that prompt a circular economy becomes more feasible. This enables the use of materials innovatively, such as upcycling plastic waste into new, valuable products or better labeling existing products for post-consumer processes. The future scenarios were developed by researching how plastic waste is dealt with today and from the expert interview. They were constructed based on previous research on current recycling methods and the insight from the interview.

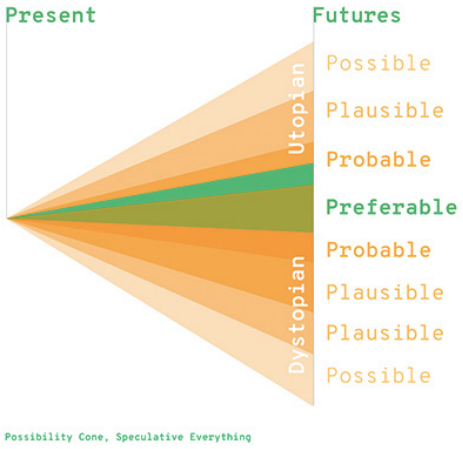


Fig 4. Possibility core. Anthony Dunne & Fiona Raby (2013)

4.4 Sketching

Sketching refers to the development of exploring concepts and designs (Rojas, 2024). It is a fundamental aspect of the design process because it lets designers explore and communicate their ideas visually. It serves as a medium for rapidly generating and iterating on concepts, allowing for quick visualization of different ideas. Rojas (2024) implies that sketching is an organic approach to problem-solving, providing a flexible platform and innovation in the design process. This was managed using pen and paper.

4.5 Prototyping

Camburn et al. (2017) describe prototyping as the design process that turns ideas and concepts into tangible representations for testing and evaluation. Prototypes are meant to simulate the final products for analysis before committing to produce an actual product. By creating physical or digital prototypes, designers can gather feedback, identify flaws, and make improvements before finalizing a product. Camburn et al. (2017) explain that prototyping allows designers to physically explore the functionality, aesthetics, and user experience, ultimately gaining insights into potential challenges and refining the design. It bridges ideas and implementation, facilitating an iterative user-centered approach that eventually leads to more successful and user-friendly products. This project used 300 gsm drawing paper, tape, scissors, a utility scalpel, a ruler, and a cutting board to create prototypes based on concepts chosen for further development during sketching.

5. Results & Analysis

5.1. Results of MDD

5.1.1 Material understanding

The material understanding resulted in a table shown in Figure 5, showcasing ABS with comparable competing materials in similar lines of application that is based on Thompson's (2017) materials source-book, such as polycarbonate (PC), polyamide (PA), polypropylene (PP), and polyethylene (PE). It shows that ABS plastic requires the lowest energy; however, the raw-material cost is more significant.
The insight from the material understanding is that ABS has a crystal melting temperature feasible for designers with limited tools, allowing for tinkering and exploring forms for possible prototypes, as designers may test various thicknesses and bends on the material. Additionally, how can the project enable products made of ABS to be effectively identified and recycled?

Material	Density (g/cm^3)	Crystal melting temperature (°C)	Elongation (%) Min-max	Strength-to-weight (length of the equivalent strip before material fails under its weight)	Cost Raw price material ($/kg)	Production Energy production kJ/kg	Recycling End-of-life recycling
ABS	1.0-1.1	190	5-25	1.0	2	60	Recyclable & identified Identified by code #7. It is not typically collected from mixed waste.
PP	0.9-0.91	176	16	1.0	1	80	It is readily recyclable but has yet to be widely collected. Contains harmful chemicals subject to restrictions, such as bisphenol A (BPA)
PC	1.2	230	60-100	1.1	3	100	Identified by code #5. High embodied energy Effectively recyclable and separated.
PE (HDPE)	0.95-0.97	137	10-1000	1	1	80	Identified by code #2. It is effectively recyclable and separated.
PA	1.1-1.2	210-220	40-60	1.3	2	125	It is recyclable and identified by code #7. It has relatively high embodied energy. However, recycling is challenging due to the wide range of types and mix of ingredients.

Fig 5. Material understanding table. Louis Diep (2024)

5.1.2 Tinkering

The process resulted in 5 successful samples using different temperatures between ABS crystal melting temperatures. The first samples (A & B) that are shown in Figures 6 and 7, explored if there is a difference between molding materials from granules (A) and scrap directly from the waste (B), which resulted in little to no difference. The second sample in Figure 8 shows the results of increasing the temperature to 140 Celsius. The central insight from the tinkering is that higher temperatures, such as 160-170 Celsius, achieve smaller thicknesses and are easier to reshape - as shown in Figures 9 and 10. Tinkering with temperatures lower than 120 Celsius was redundant as they were not hot enough to bind and create samples, and tinkering with temperatures higher than 180 Celsius risks melting ABS, releasing harmful fumes that can potentially harm consumers and designers. In the case of sample F in Figure 11, the sample contained a mixture of ABS and PP, which caused it to melt and result in an abnormal sample. This sample has a significantly thicker thickness of around 0.7-1 mm and was far too brittle to work with. This is why this project has limited the research to one type of plastic.

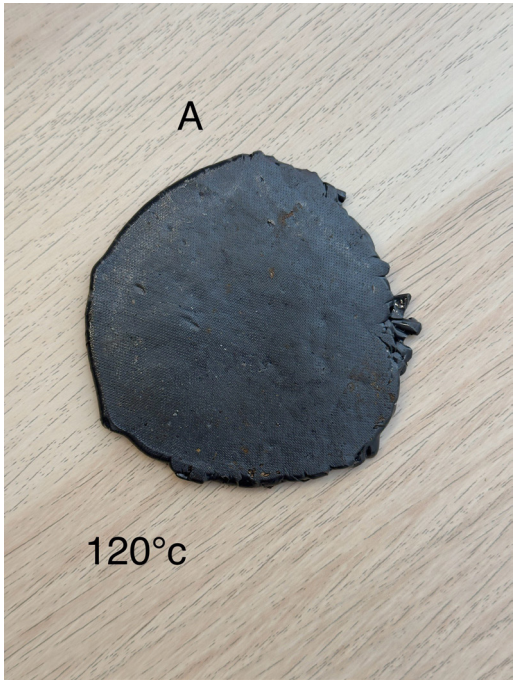


Fig 6. Sample A.Louis Diep (2024)

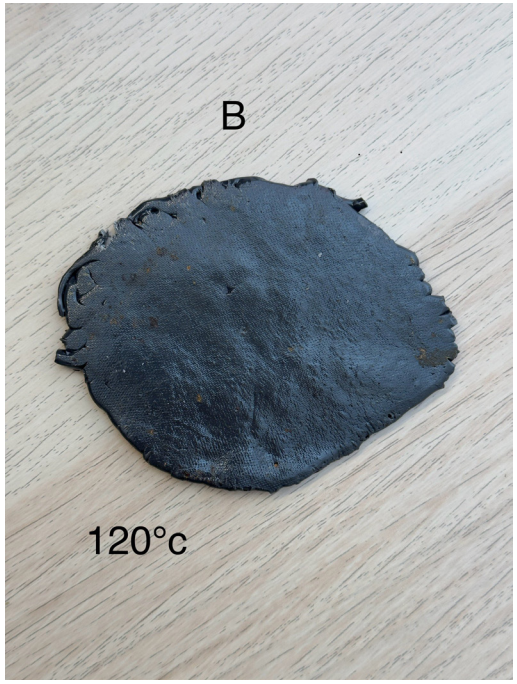


Fig 7. Sample B. Louis Diep (2024)

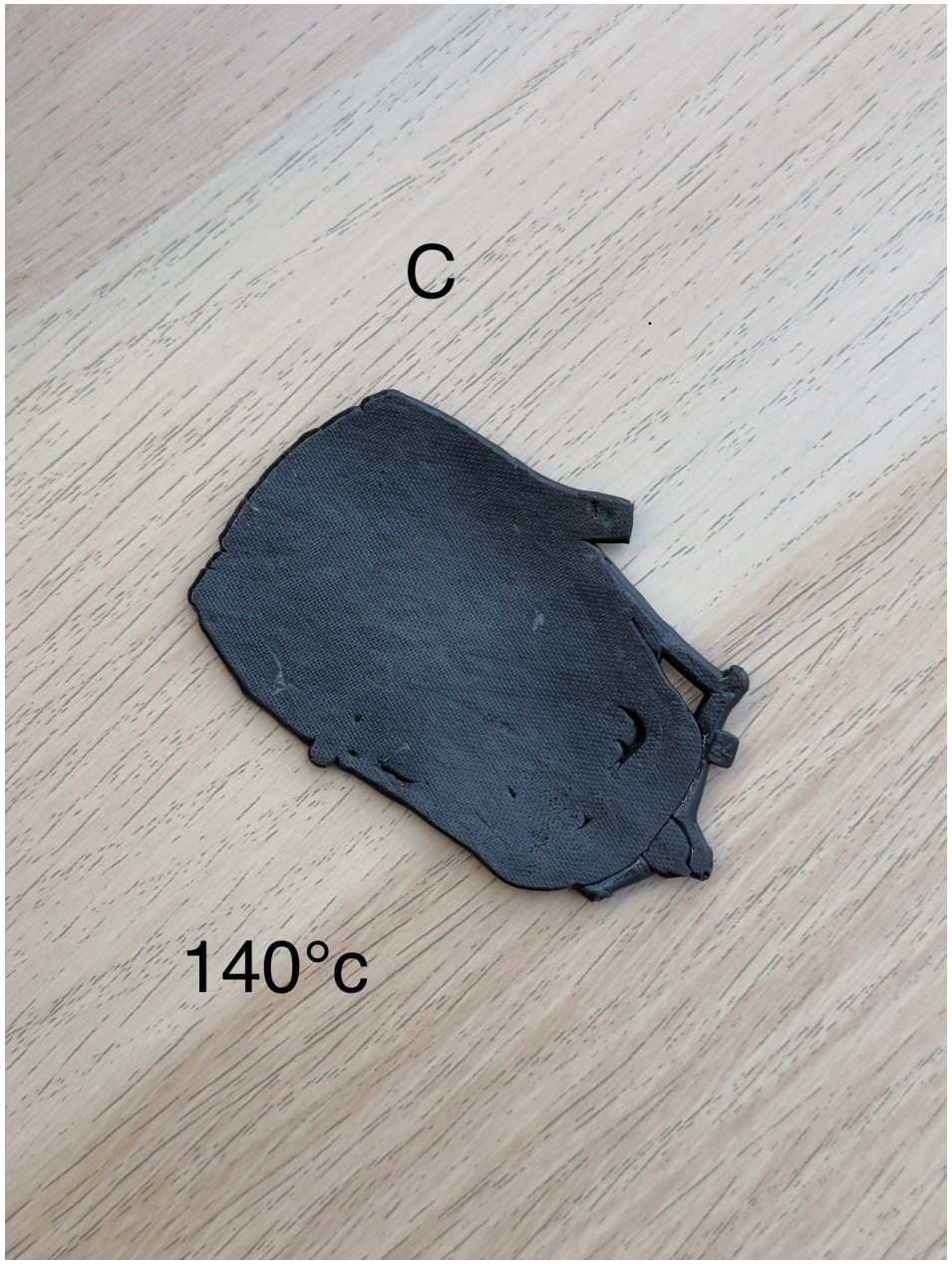


Fig 8. Sample C. Louis Diep (2024)

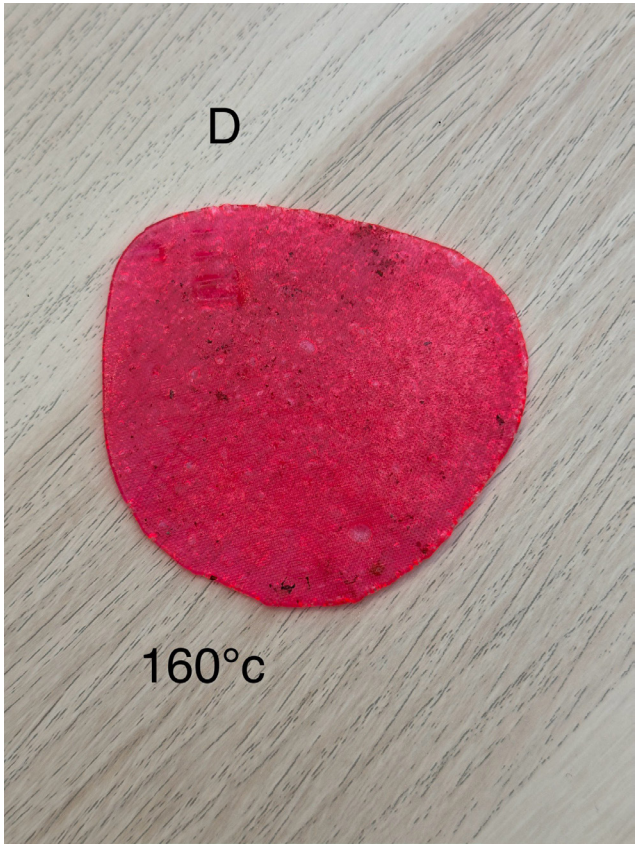


Fig 9. Sample D. Louis Diep (2024)

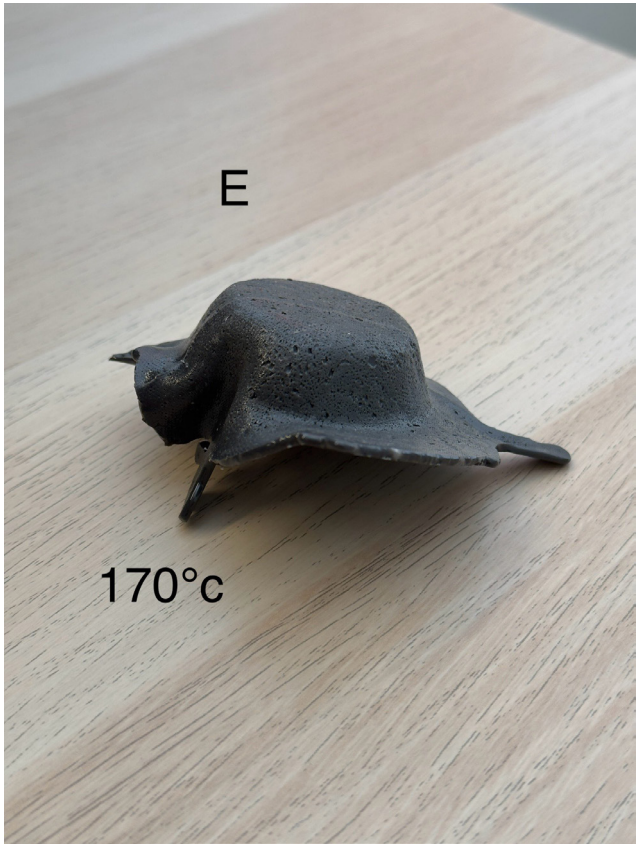


Fig 10. Sample E. Louis Diep (2024)

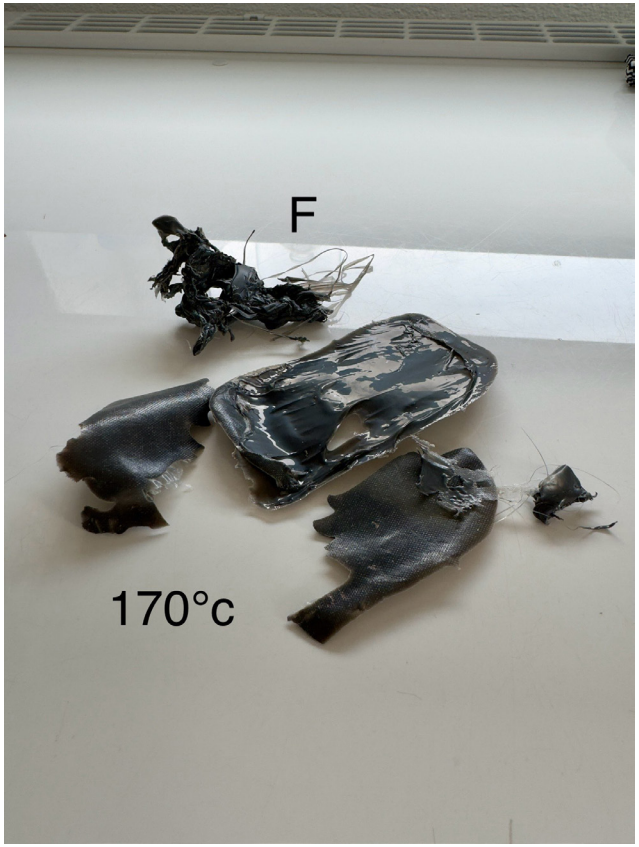


Fig 11. Sample F. Louis Diep (2024)

5.1.3 Benchmarking

The benchmarking process resulted in a table showcasing the samples made previously. Figure 12 reveals that the thickness of the material is related to the inputted temperature, as lower temperatures, such as 120 Celcius, yield a thickness of around 6 mm. In contrast, temperatures higher, such as 140-170, make a thickness as thin as 1.9 mm. The project decided to continue using sample E as a reference to design and construct the final product as the sample has a thickness that is easy to mold and cut.

Sample	A	B	C	D	E
Temperature	120	120	140	160	170
Texture	Smooth	Smooth	Smooth	Smooth	Smooth
Thickness	6mm	5.5mm	2.2mm	2mm	1.9mm
Natural color	revealed	revealed	revealed	revealed	revealed
Imperfection	low	low	low	Medium	low
Roughness	low	low	low	low	low

Fig 12. Benchmaring table. Louis Diep (2024)

5.2 Results of the Interview

The interview was conducted with an expert from Swedish waste management, Sysav, who gave insight into current recycling methods and the following challenges.

How is plastic waste separated today?
- There are multiple ways to separate plastic. Individuals may visit recycling centers where they collect hard and soft plastic. The individuals separate the plastic by hand. Some establishments also bring sortable waste to waste management. Waste management has its separation process; however, the vast majority of plastic waste goes to energy recycling through incineration. Manufacturers ‘ ABS plastic, in the form of car fenders and bumpers, is collected to explore recycling possibilities.

How is mixed plastic waste sorted?
- Mixed hard plastic goes to the waste management sorting facility, where machines sort it by identification and type so it can be recycled. Soft plastic is washed and ground into granulates.

Is the plastic waste coming to the waste management identified?
- The sorting facility identifies and labels the plastic as it arrives in large quantities and sizes, such as large HDPE or PVC pipes. These pipes can be identified by color and surface texture; for example, PVC pipes are either orange or gray and have smooth surfaces, whereas HDPE pipes are black and have wave-like surfaces. Other plastics must be identified using codes and markings, which is challenging.

Why is the majority of plastic incinerated despite the potential of being recycled?
- The incentive to recycle plastic needs to be revised due to its low value and the recycling process requiring several steps, which cost additional investment and energy. It is financially and more efficient for manufacturers to buy virgin material rather than recycled plastic. Plastic waste has to be sorted, labeled, washed, and granulated to be returned to the system. In addition, the end-life of most products is not included in the design, which makes it difficult for the product to break down and be recycled.

What sort of future scenario can be speculated regarding the use of plastic and recycling based on the knowledge of current methods and advances?
- Based on statistics from Naturvårdsverket, the usage of plastic will triple by the year 2032. More waste management facilities will be constructed as a result. The European Commission will implement regulations forcing companies to be more transparent with their products and identify them appropriately.

5.3 Results of Future Scenario

The results of the future scenario are two scenarios.

In scenario 1, the European Commission is working to meet the plastics strategy plan in 2030, aimed at reducing the environmental impact by restricting the use of only six specific types of plastic - PET, HDPE, PVC, LDPE, PP, and PS (European Commission, n.d). This regulation implies that companies will be forced to adjust their production processes to be one of the specified materials, potentially reducing plastic pollution and facilitating a sustainable approach to plastic usage within the European Union. These regulations promote environmental awareness and recycling among consumers as the products are more accessible to identify and separate. With the EU establishing a standard on sustainable products, nations outside of the EU, such as the USA, China, and Japan, with a large customer base in the EU, will be inclined to adopt similar regulations to maintain trade relationships with countries in the EU. Scenario 1 plausibly resolves the issue of mixed streams of unidentified plastic waste; however, it does not address how to manage the existing plastic with code 7.

In scenario 2, the European Commission has implemented stringent regulations requiring clear, standardized identification of plastic products to achieve a carbon-neutral and circular economy by 2030 through its plastics strategy (European Commission, n.d). This initiative aims to enhance transparency in the plastic manufacturers, facilitate recycling and proper disposal practices, and ultimately contribute to the transition toward a circular economy and environmentally friendly society. Similarly to scenario 1, countries outside the EU may be compelled to adopt these regulations to maintain their trade partnerships with the EU nations. Based on their insight and statistics, the second scenario that the expert from Sysav anticipated made the project consider how to label products so that they can be easily sorted and recycled.

The central issue of scenario 2 means to address is how to be more transparent with the product's material that society currently lacks. One solution is to design a symbol that helps indicate what type of plastic the product is and how many times it has been recycled. This resulted in two simple marks identifying what type of plastic the product is made of, such as ABS, with two arrows circling to signify that it is recyclable and another mark indicating how many times the material has been recycled, using a number with arrows circling. This creates a preferable future for society as mixed waste can be identified and sorted, potentially reducing plastic pollution and landfill.

5.4 Results of Sketching

The sketching process resulted in various sketches with different themes. The first phase of the process resulted in iterations of products that ABS is typically used for, such as helmets, toys, and pots, as seen in Figures 13, 14, & 15. These were iterated on to present the potential of upcycling old products into new ones. Sketches of similar value were also iterated to widen the horizon and explore different design styles. However, after reflecting on the sketches of the first phase, the second phase moved on to iterating products to ABS, which is not usually applied, such as furniture, to mirror the usage of ABS, as plastic is a mundane part of modern life.

The second phase aimed to break the image of ABS products creatively by interplaying functionality and aesthetics. This resulted in sketches of objects, such as sketches of benches, chairs, tables, and lamps, as seen in Figures 15, 16, 17, & 18. Designing a product perceived as high value and highly appreciated fits the concept as it prolongs the product's life cycle due to care and maintenance. After receiving feedback from the handler, it was suggested that the product's material be exposed to show the texture of the surface and exhibit the story of how the material has been processed, which works well as the project's goal was to explore how plastic waste can be upcycled to create new products.

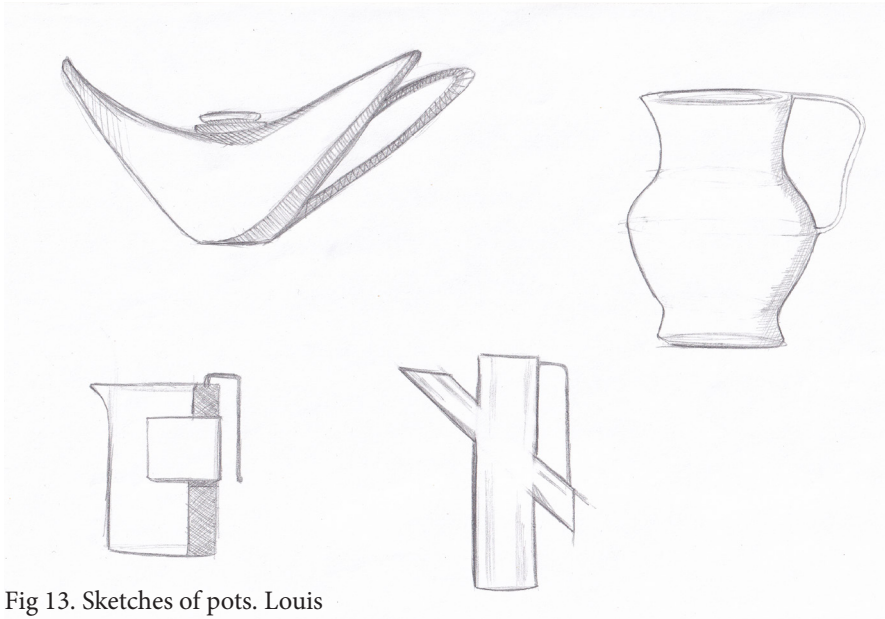


Fig 13. Sketches of pots. Louis Diep (2024)

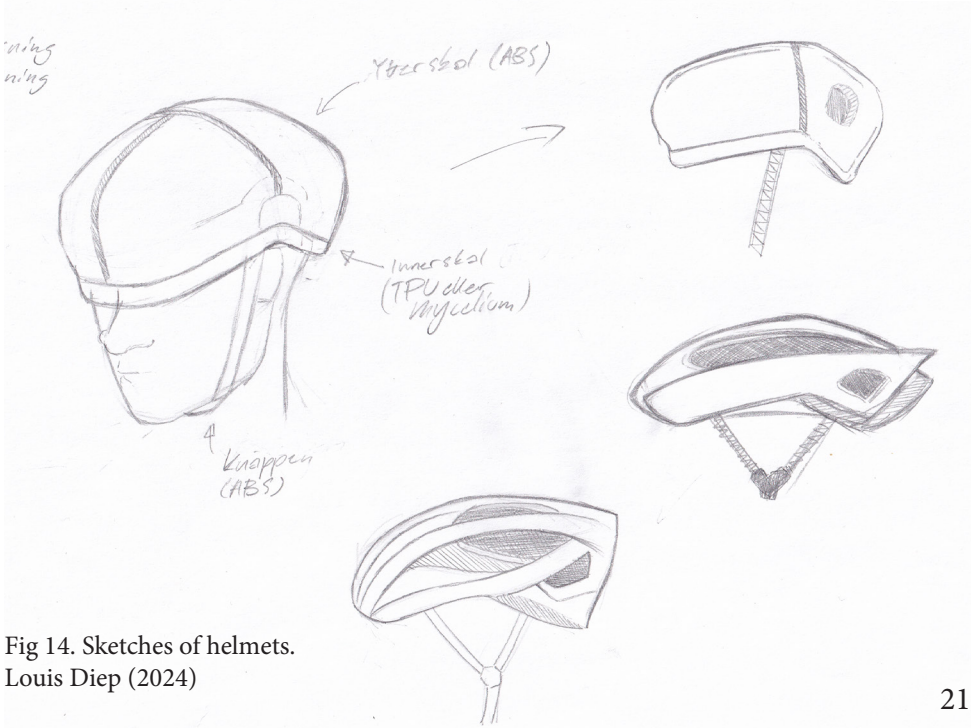


Fig 14. Sketches of helmets. Louis Diep (2024)

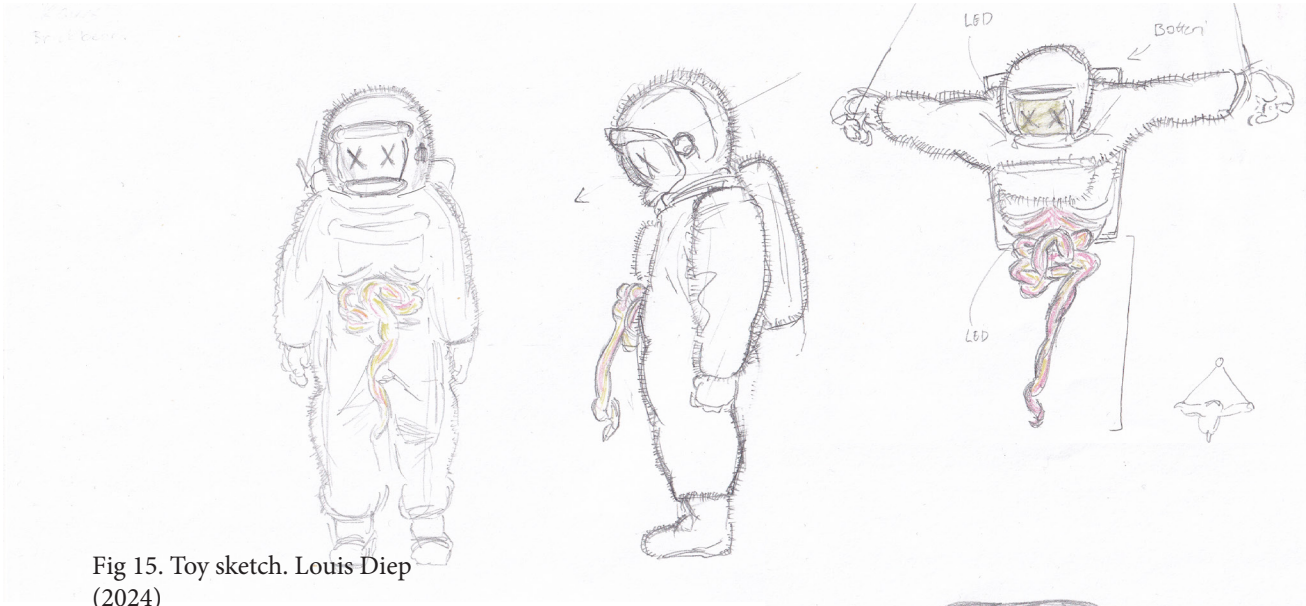


Fig 15. Toy sketch. Louis Diep (2024)

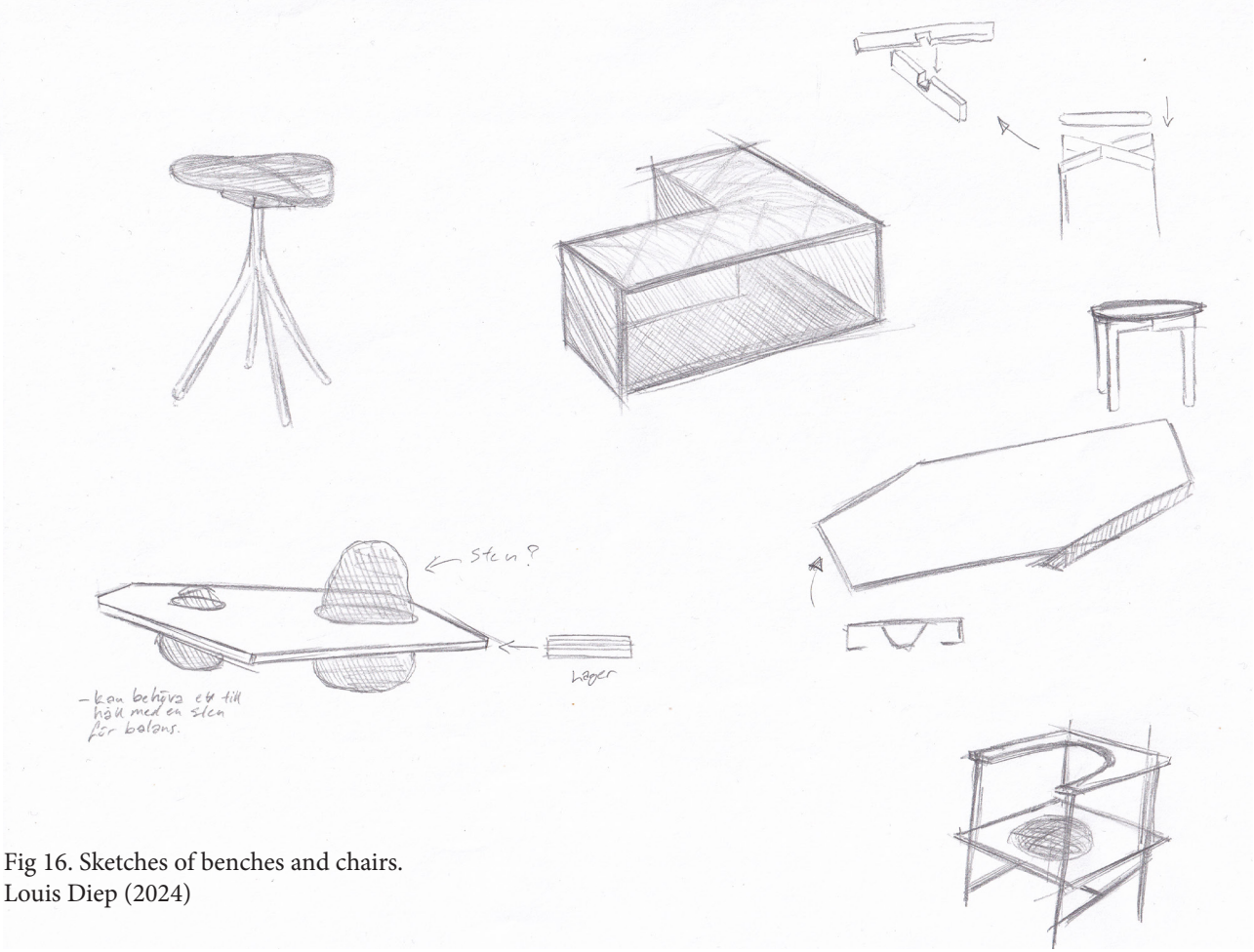


Fig 16. Sketches of benches and chairs. Louis Diep (2024)

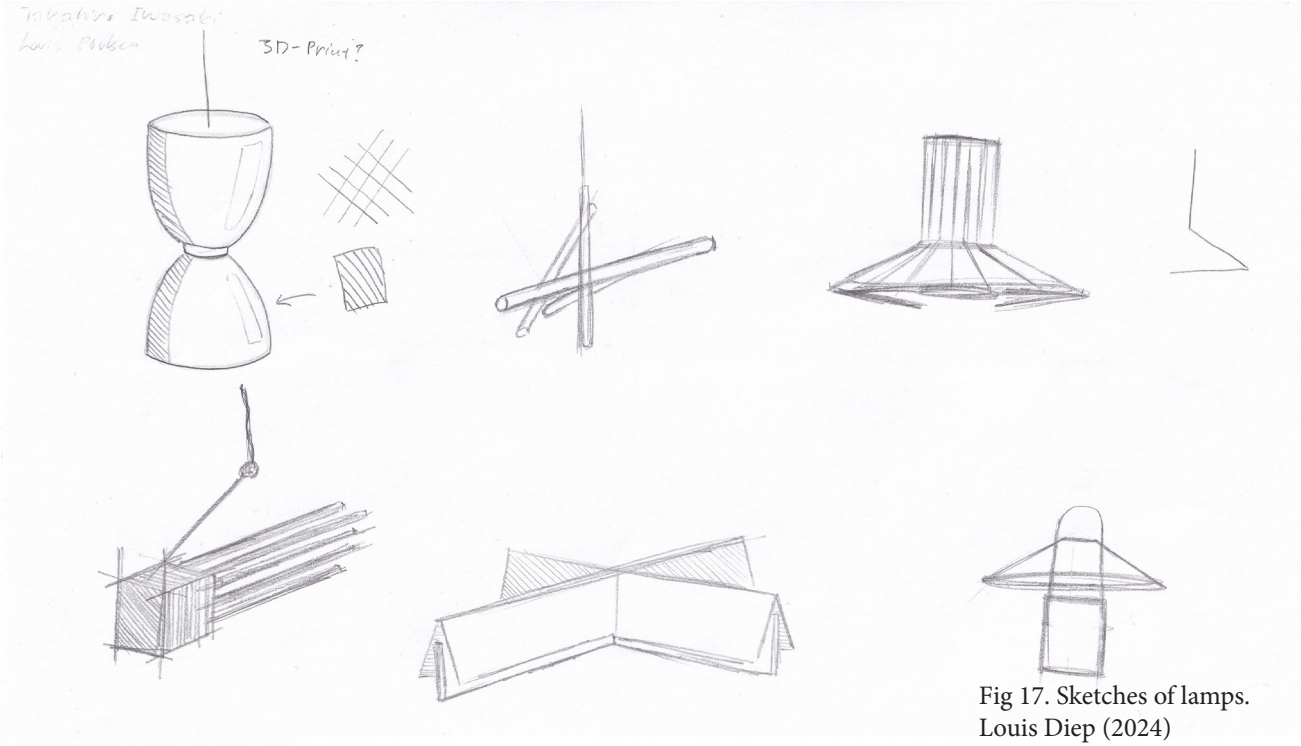


Fig 17. Sketches of lamps. Louis Diep (2024)

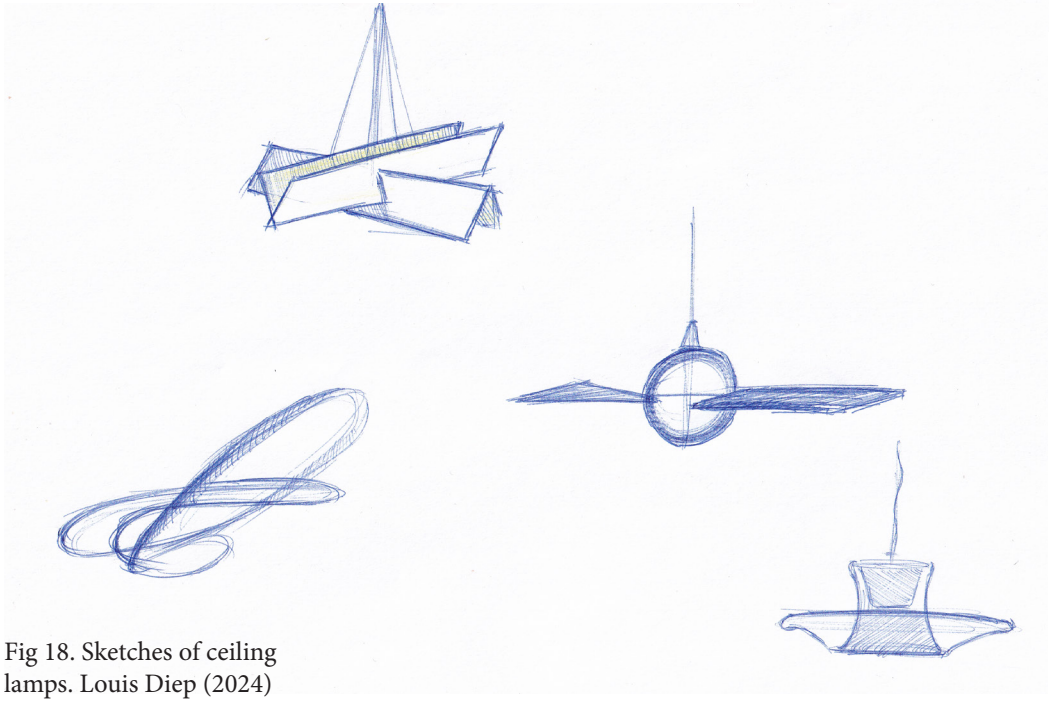


Fig 18. Sketches of ceiling lamps. Louis Diep (2024)

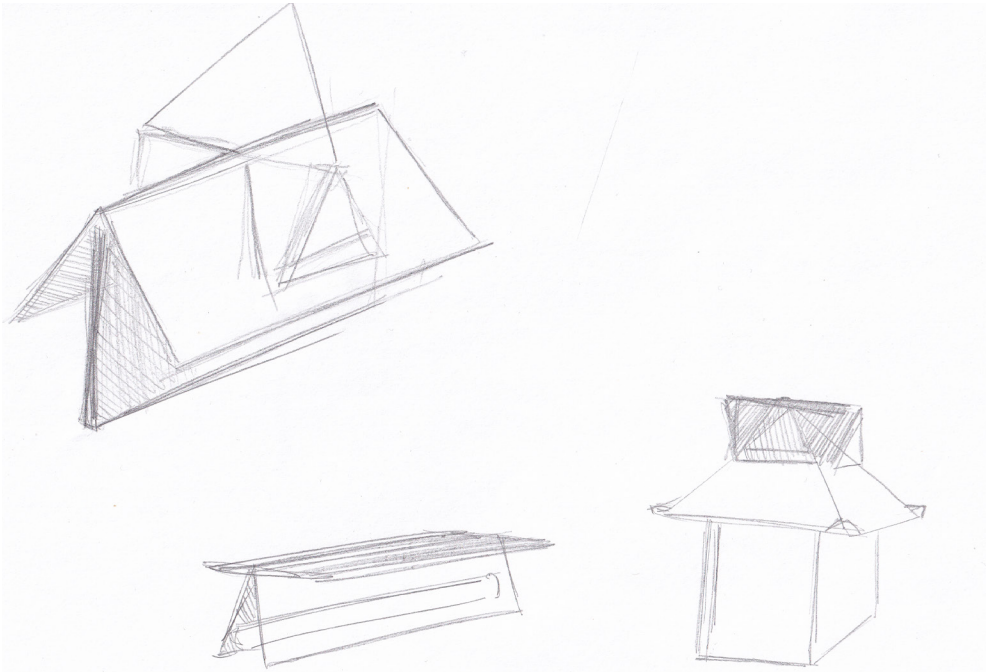


Fig 19. Sketches of ceiling lamps. Louis Diep (2024)

The feedback and result of MDD suggested designing a product that refrains from active handling. The project decided to work on a hanging lamp seen in Figures 19 & 20, as lamps are not actively handled and have interplay with the form and lighting. This allowed the product to tell a story through its exposed material. The sketches were refined, resulting in hanging lamps inspired by pagodas and modern Scandinavian design. The sketches gravitated toward these motifs because they are typically associated with exposed materials such as wood, metal, and stones. In addition, they are respected and highly valued due to their long history, appreciation, and care from people.

The continued sketches were chosen after considering their feasibility and whether they aligned with the project's motif and overall aesthetic. The lamp's design seen in Figures 21 & 22 aims to convey serenity with long arches similar to the roof of pagodas but takes a minimalistic form familiar to Scandinavian design that values simplicity.

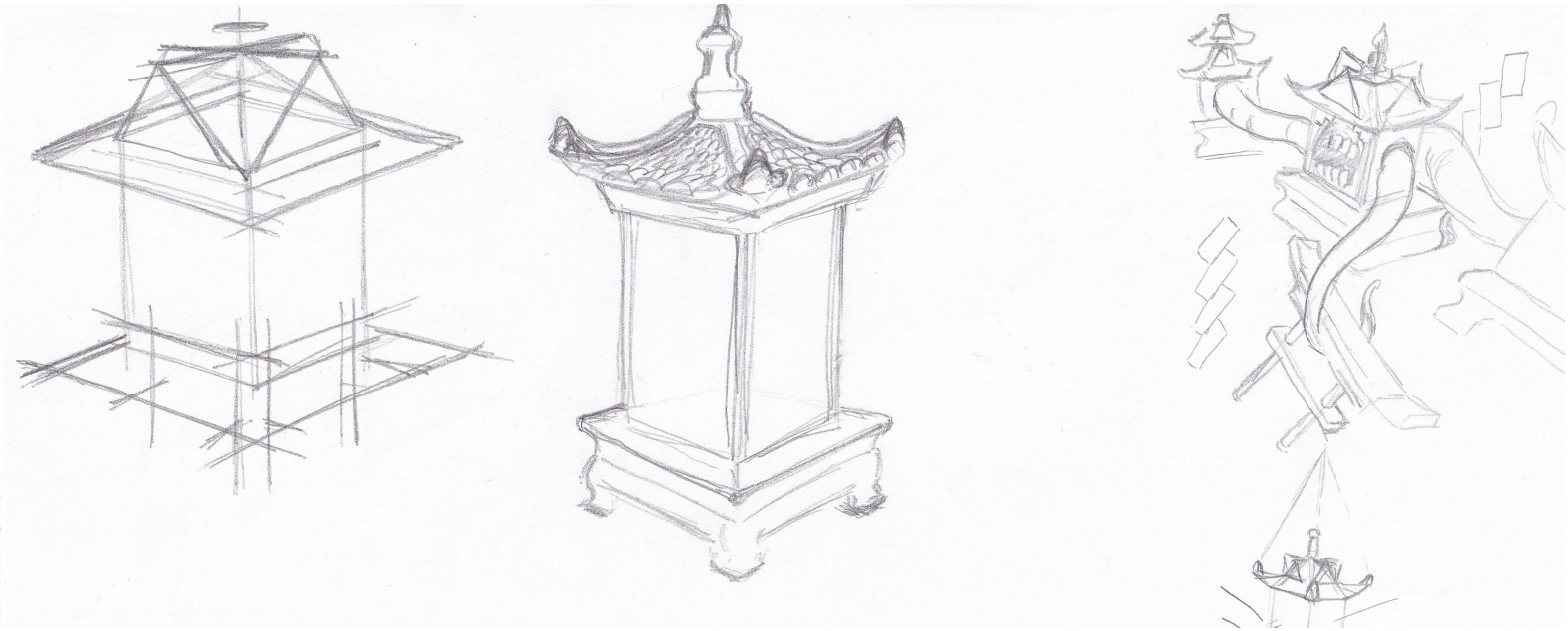


Fig 20. Sketches of pagodas. Louis Diep (2024)

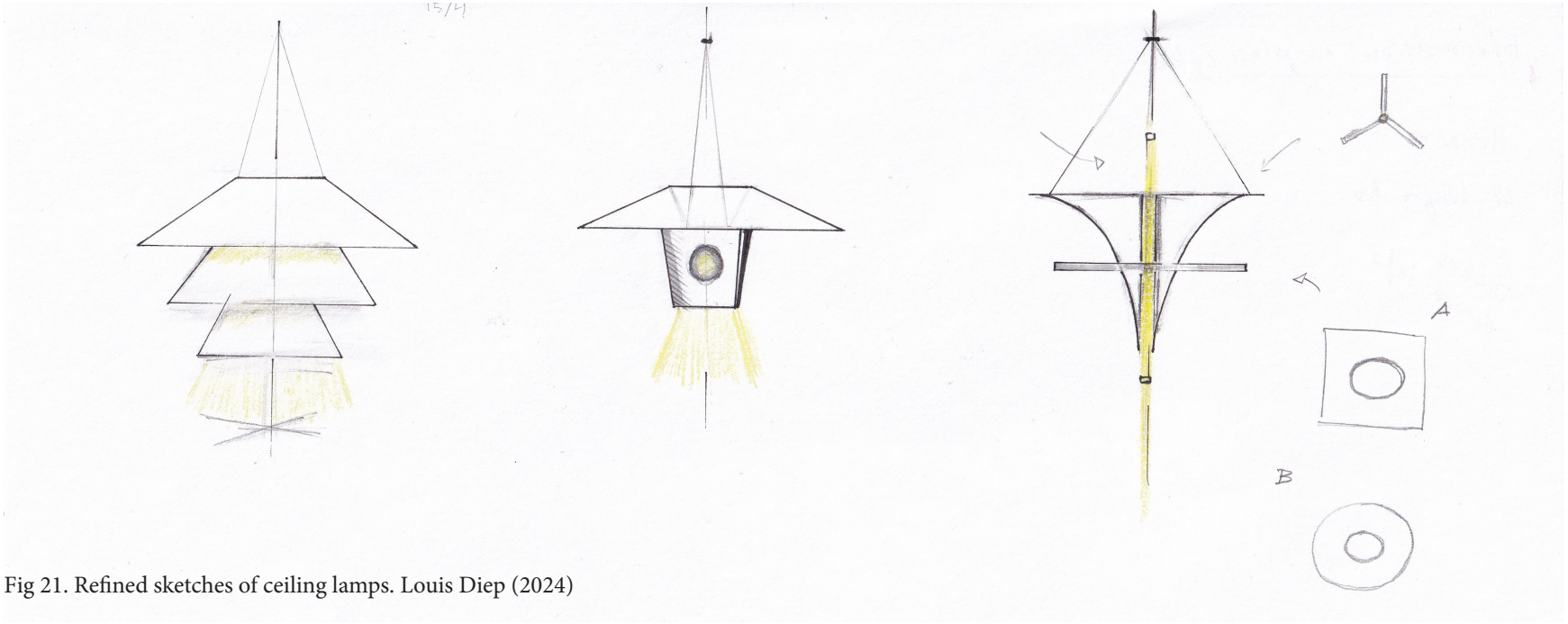


Fig 21. Refined sketches of ceiling lamps. Louis Diep (2024)

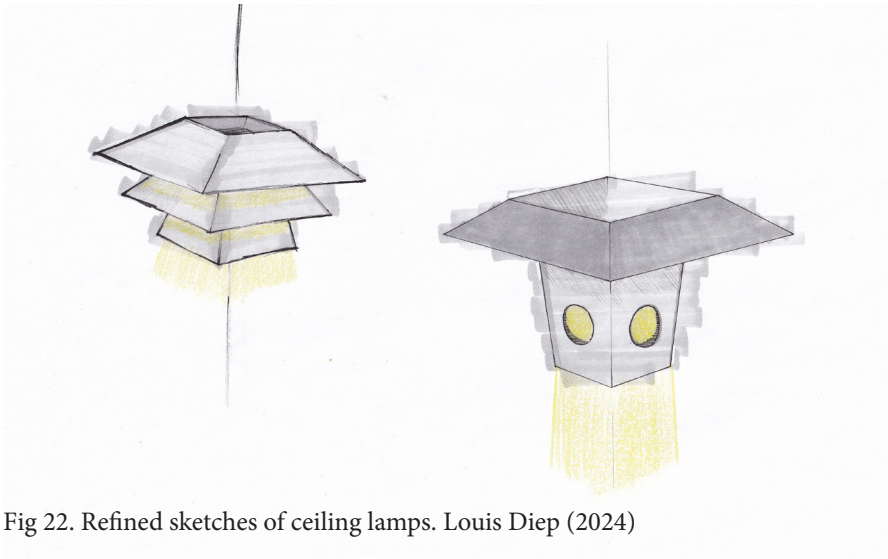


Fig 22. Refined sketches of ceiling lamps. Louis Diep (2024)

5.5 Results of Prototyping

The prototype resulted in three simple lamp prototypes. The design of the lamps was chosen because they are feasible and resonated with the concept that was explored, in this case, pagodas and Scandinavian design. Due to the construction and material, the prototypes are flimsy; however, they effectively convey the three-dimensionality of the design and aid in analyzing the construction aspect. Although they may not be very sturdy as displayed in Figures 23, 24, and 25, their ability to bring the designs to life is beneficial for further development and refinement. The prototypes aided in exploring how to attach the lightbulb to the lamp and how the part will be connected, screws or adhesive. A revelation discovered from prototyping is that the product can effectively be built and packaged compactly, similar to how IKEA furniture and products are packaged displayed in Figures 26 & 27.

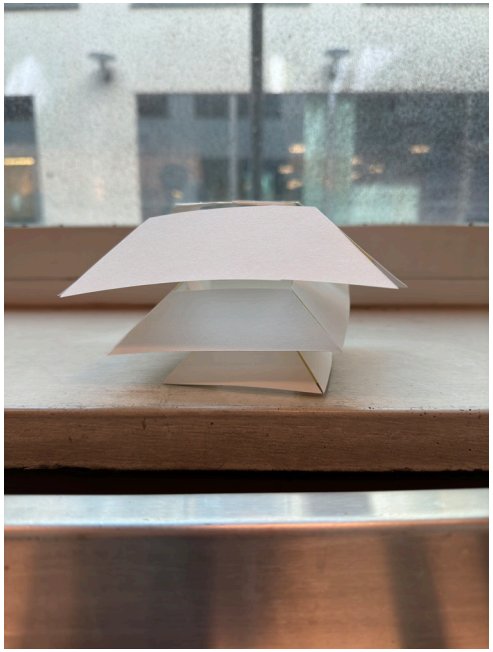


Fig 23. prototype. Louis Diep (2024)



Fig 24. prototype. Louis Diep (2024)

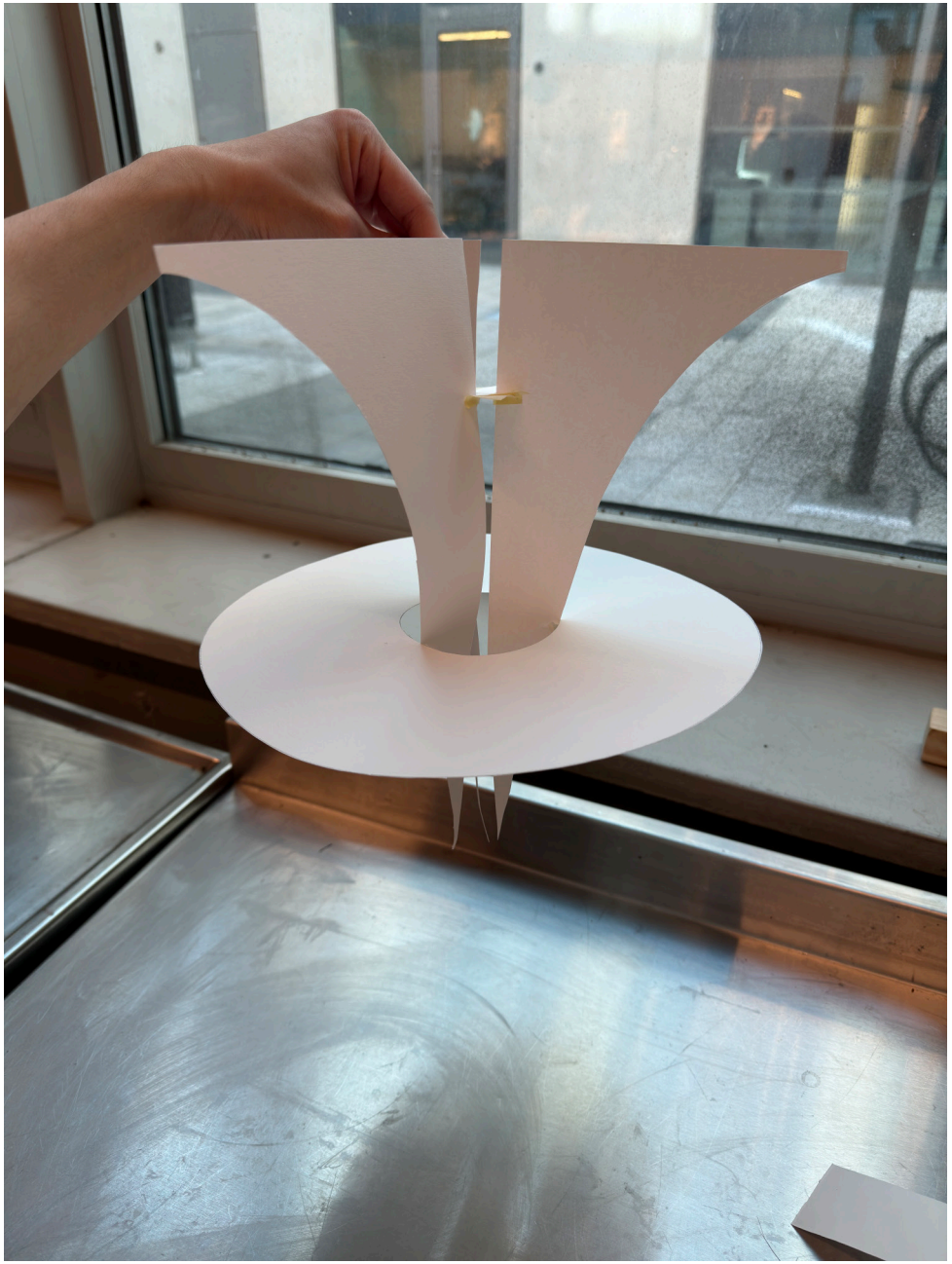


Fig 25. prototype. Louis Diep (2024)

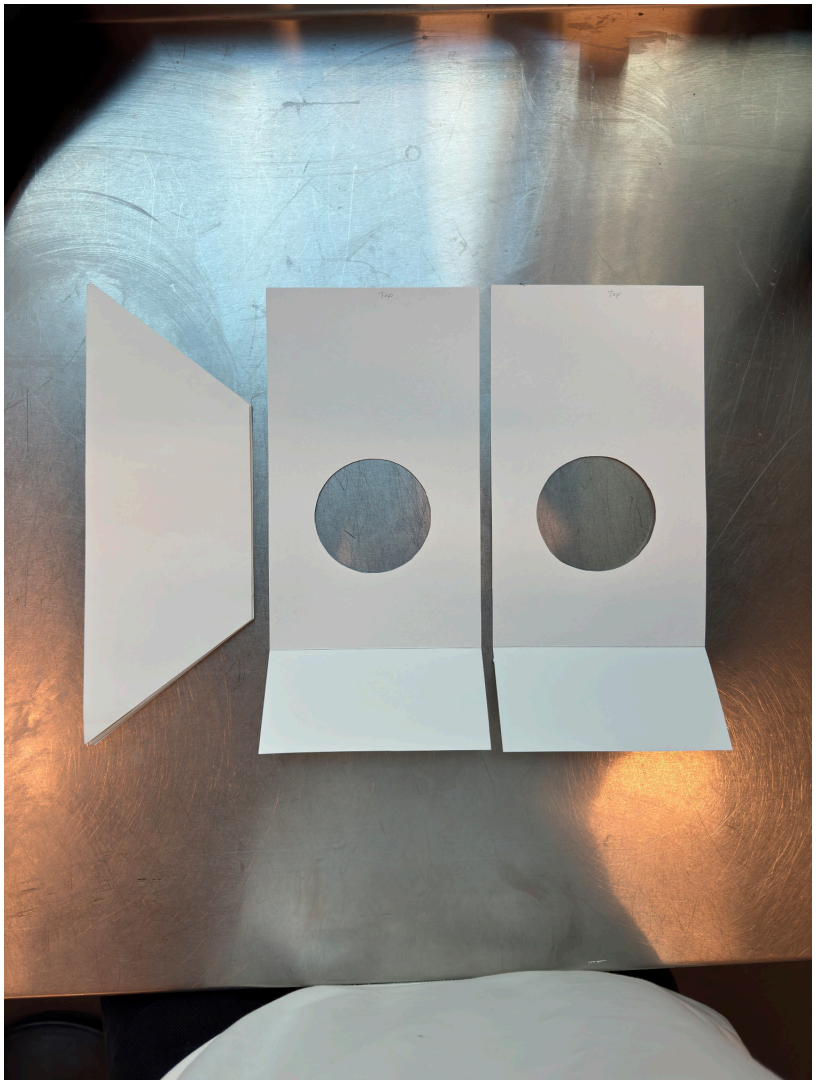


Fig 26. prototype folded. Louis Diep (2024)



Fig 27. prototype folded. Louis Diep (2024)

6. Final concept

The final concept for the product is a hanging ceiling lamp based on the third sketch and the third prototype. The design decision was chosen because of its intricate build and ambiguous form that combines tranquility from pagodas with the modern, simplistic Scandinavian design. Figure 29 exhibits that parts of the lamp, such as the main body and the lightbulb holder, will be made of recycled ABS plastic recovered from various scraps, such as broken helmets or car bumpers. The exposed imperfections and texture seen in Figure 28 reflect the journey of the material and the work of MDD, as they highlight how the project has worked with ABS through tinkering and benchmarking. Figures 30, 31, 32, and 33 display the lamp from different perspectives to show how it looks from different angles.

When turned into a lamp, parts using ABS will be marked with a distinct label that informs what sort of material and the number of times the part has been recycled starting by the designer since the designer may not be able to trace the number of times the material has been recycled during its time by the manufacturer.
Additionally, it is challenging to distinguish between products using virgin materials and recycled products, as most products use a mixture of virgin ABS and recycled materials, between 30-70 percent (Thompson, 2017). These labels are created by heat engraving, simply the abbreviation of the material that consumers may know of, in this case, ABS, and a number with a circle (seen below) to display the amount of recycle loop the material has gone through. The label is built on the second future scenario that projects the European Commission's goal to reach a carbon-neutral society by facilitating a circular economy through material recyclability and circular design.



Fig 28. product. Louis Diep (2024)



Fig 29. product. Louis Diep (2024)

When a part is broken, the consumer may return it in exchange for a new one; meanwhile, the broken part will be recycled and labeled with a new number indicating that it has gone through a recycle loop. The marking will be placed inside the lamp, which can be concealed when displayed but easy to find when opened. Ultimately, when the ABS has been recycled through too many end-of-life cycles and starts showing a loss in the build quality or color, the ABS is sent to waste management, where it can be recycled through incineration.



Fig 30. product. Louis Diep
(2024)



Fig 31. product. Louis Diep
(2024)



Fig 32. product. Louis Diep
(2024)

7. Conclusion & discussion

How can end-of-life material from plastic waste be used to create products within product design?

This study has explored how to use end-of-life plastic waste from ABS by upcycling and repurposing scrap into ceiling lamps. An effective method to upcycle ABS with a low threshold as a designer or a consumer interested in upcycling–is to use a contact grill and set the temperature depending on the desired material thickness, around 160-170 to yield thin material thickness and 120 Celsius to get thick material thickness. This allows designers and consumers to safely experiment with the material and shape different forms without exposure to harmful fumes.

The fundamental problem concerning recycling waste lies in adequately identifying products and materials, specifically plastic, which results in the vast majority of the waste being incinerated rather than recycled (Sysav, n.d). Plastic labeled with identity code number seven ends up in large mixtures with other types labeled with code number seven, which are too demanding to separate and recycle. The mixture must be sorted out, washed, and finally granulated to reach the same position as virgin material, which costs less as it does not have to go through the same process—as stated by experts from waste management interviewed in this project. Manufacturers' demand for recycled materials drops as it is less cost-effective and requires more facilities to process plastic waste. However, the expert did not have a number or percentage on the current status quo of how cheaper it is to source new virgin material compared to sourcing recycled material. This study offers an alternative by creating a product connected to an open-loop cycle, which holds the product in a longer life-cycle where parts are recycled rather than having a linear life where it ends up discarded. Intrusive and straightforward labels that indicate what sort of material the product is made of and how many times it has been recycled encourage the consumer to separate—sorting waste effectively and helping manufacturers and waste management to recycle more material instead of incinerating the waste and legislating regulations that force manufacturers to be transparent with their products and standardized labeling of recycled material help to facilitate environmental awareness among consumers. It hinders an excessive landfill, incinerates waste of value, reduces plastic pollution, and promotes a circular economy.
and promotes a circular economy.

If there had been more time and resources, the project could have tried different textures on the contact grill to obtain other surfaces or used extrusion molding to produce parts instead. The project could have also explored more advanced methods to upcycle plastic, such as vitrimerization or the fabrication of nanocomposites. However, these methods have recently developed and are considered emerging, according to Balu et al. (2022), who state that they can potentially be applied to future production. Upcycling is still a developing approach to design for sustainability that has yet to reach its full potential; however, the boundless creativity and potential make upcycling feasible and utilized among designers.

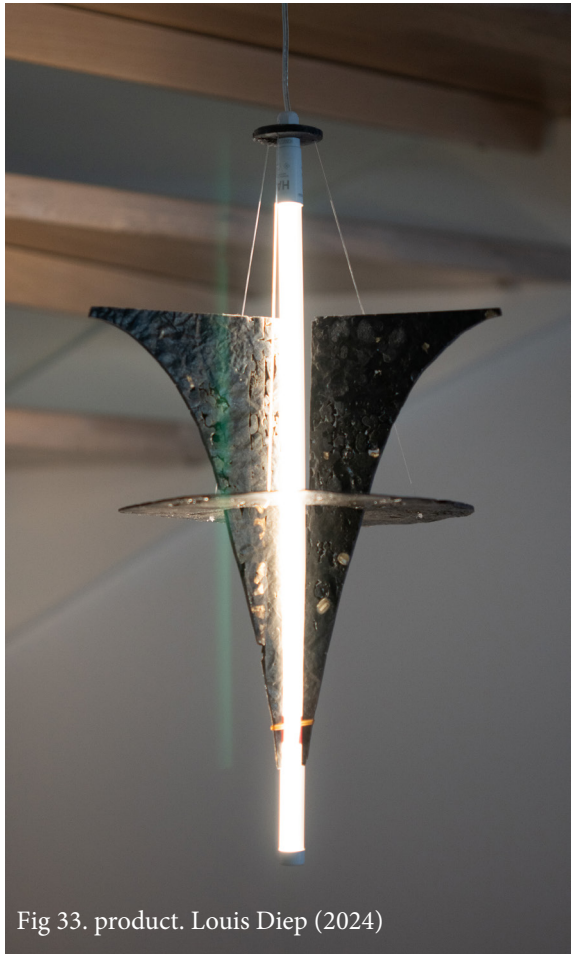


Fig 33. product. Louis Diep (2024)

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9. Appendix

9.1 Appendix. Interview

How is plastic waste separated today?
How is mixed plastic waste sorted?
Is the plastic waste coming to the waste management identified?
Why is the majority of plastic incinerated despite the potential of being recycled?
What sort of future scenario can be speculated regarding the use of plastic and recycling based on the knowledge of current methods and advances?

9.2 Appendix. Models

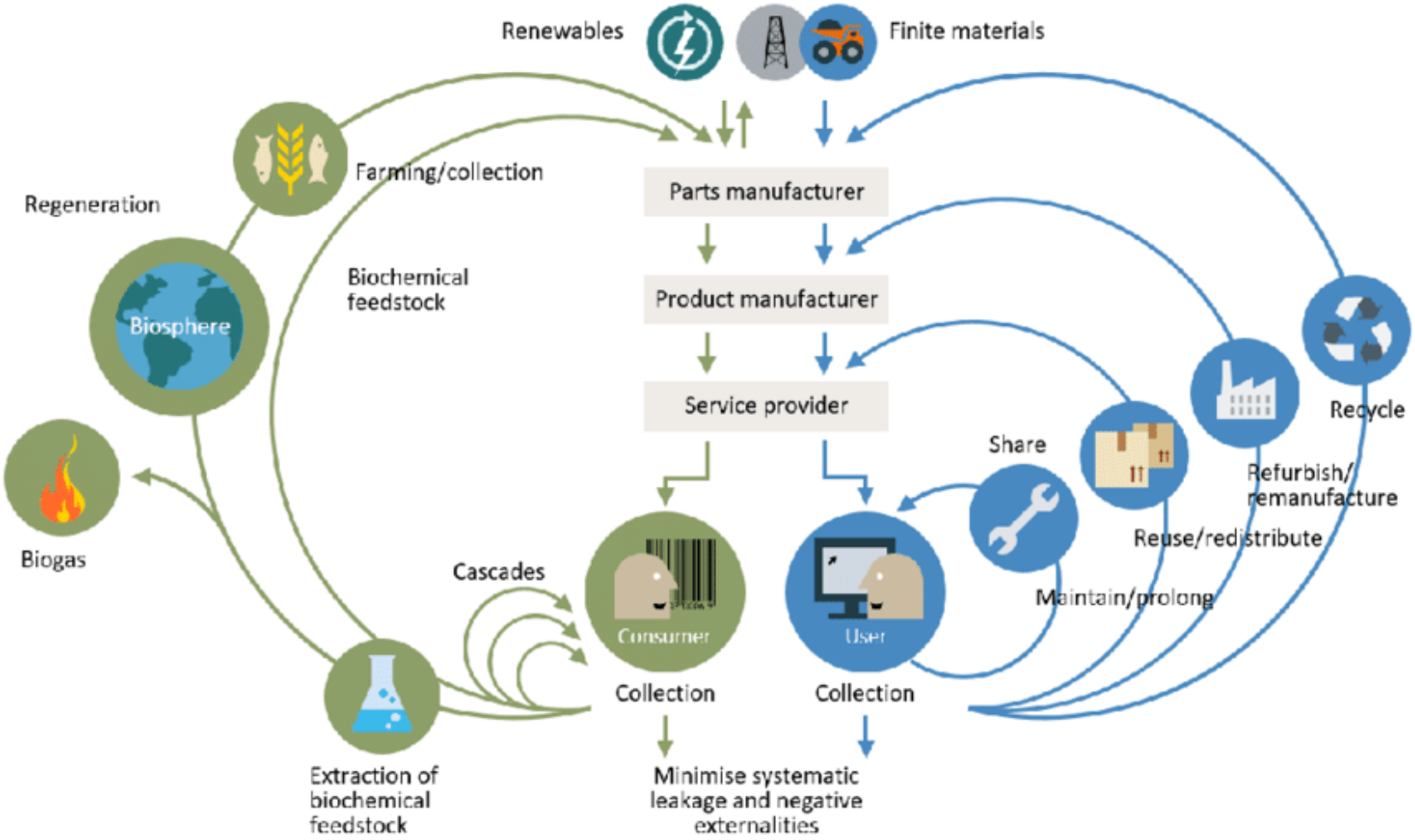


Fig 1: Ellen Macarthur Foundation 2019

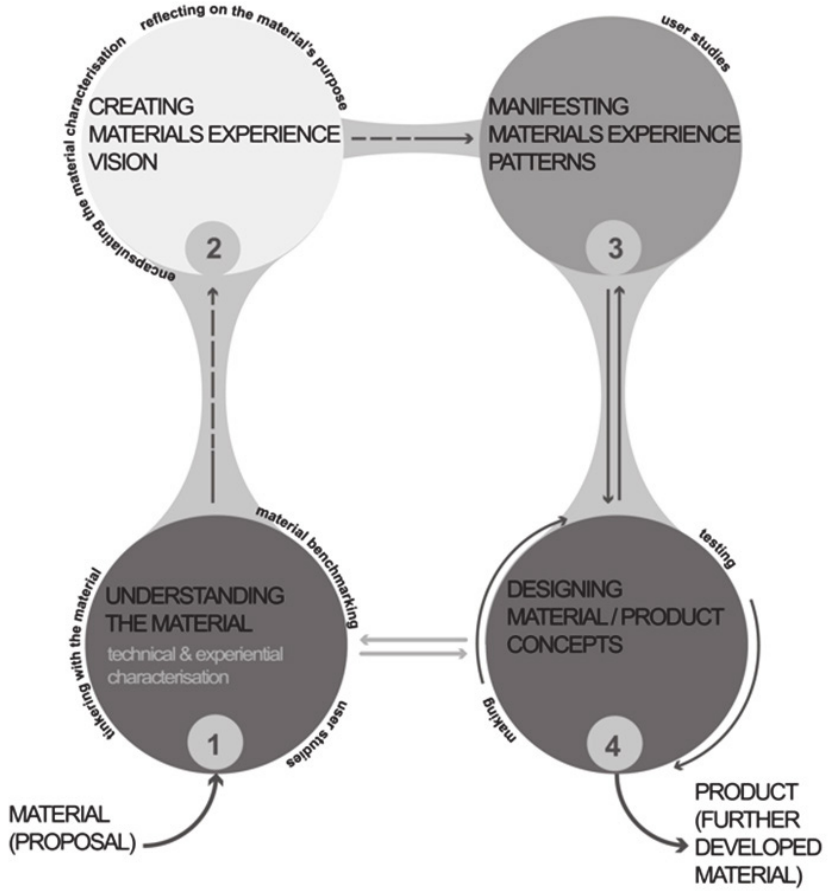


Fig 2. Material Driven Design (MDD) method. Karana et al. (2015)